

Stock analysis of S&P 500, Nikkei 225 and FTSE 100

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1 Introduction

1.1 Domain of interest

The report put Finance as a core domain of the topic and demonstrated analysis focused on the significant influenced companies, which were listed up in the representative stock price indexes in the United States (US), the United Kingdom (UK) and Japan.

1.2 Target audience

The report was targeted for stock traders especially taking short-term position, who aims to hold stocks less than a year and earn capital gain rather than income gain from tradings. Since the analysis focused on comparisons between companies included in the indexes and showed the popularity and tread at a specific point in time without fluctuation from the past and future forecasting.

1.3 Research Objective

The market capitalization of listed companies in the stock markets are fluctuated and its orders are changed depends on its stock price. To identify the companies and sectors attractive to traders, researched and visualized features of those from major indexes in US, UK and Japan stock markets. The research project selected following indexes because these indexes cover a majority part of market capitalization in each market and are composed of excellent performance companies called as 'blue-chip' [1].

- Standard and Poor's 500 Stock Index (S&P 500), covers US market
- FTSE 100, covers UK market
- Nikkei Stock Average (Nikkei 225), covers Japan market

Trustnet (2025) mentioned that S&P 500 is the most widely recognized stock index worldwide, FTSE 100 composed of the largest companies in London Stock Exchange by market capitalization, and Nikkei 225 provides insight into Asia Pacific economic trends. These indexes are influential for traders all over the world.

In addition, to identify areas which have accumulated wealth, demonstrated the geographical views distribution of companies in each country based on the indicator of the corporate size.

1.4 Short Summary

In the real world stock tradings, traders eager to seek a favored trend and invest companies, which would grow more or paying enough amount of dividends. The report focused on S&P 500, FTSE 100 and Nikkei 225 at June of 2026, and demonstrated top-ranked companies by market capitalization and sectors' features incorporating perspective of dividends. in each market. In US market, technology sector and semiconductor companies were strong such as NVIDIA, Apple, Alphabet, Microsoft, Amazon, Broadcom and Micron. The UK market was relatively old-economy but companies could pay high dividends. In Japan, technology sector

was favored but industrial companies, such as Toyota, Hitachi, Shin-Etsu Chemical, Tokyo Electron and Advantest were still strong and keep its presence. Finally, finding local excellent companies, which are located other than megalopolis areas, such as New York, Los Angeles, London and Tokyo by viewing a geographical dashboard.

2 Research Question

Fama and French (1993) described monthly returns on stocks were regressed on the return to a market portfolio of stocks and mimicking portfolios for firm size, book-to-market equity. In other words, cooperate size and market capitalization are indicators for evaluating a risk of investment to an individual company.

Cavaglia et al. (2000) pointed the change from the past research result, which was the country factors dominated industry factors, to the industry factors had been growing and dominate country factors for determining equity returns. Diversification across industries provided greater risk reduction than diversification across countries. In addition, H. Hong et al. (2006) researched the return on industry portfolios, which were retail, services, commercial real estate, metal and petroleum, and they found the correlation between an industry's predictive ability and the tendency to forecast indicators of economic activity for US stock markets.

Baker and Murgler (2004) researched the role of dividends for putting a share price. Dividends were highly relevant to a share value, but in different directions at different times. Alternatively said that there were two types of investors who preferred dividends and who disliked dividends and investors attitudes toward dividends have changed over time.

Chang et al. (2020) focused on the role of institutional ownership to the impact on corporate social responsibility (CSR) decisions, especially local long-term institution. The team pointed the an institutional ownership is an important governance mechanism that influences managerial decisions. In addition, local long-term institutional investors value benefits from CSR more because they also community stakeholders and geographic proximity helps them to better assess soft information embedded in CSR activities. Investing activities were different between local and non-local institutional ownership.

Based on the above past research results, 'Corporate size and market capitalization', 'Industry factors', 'Dividends' and 'Local and non-local' were key words for this research and generate useful insights for traders. To build a robust portfolio, which is selected from attractive companies in the market, and to make a buy / sell decisions, needed to see the current status by using the following questions incorporating the key words:

1. What companies have drawn investment money from investors in the market, by scaling its market capitalization.
2. What sectors were favored for investment in the market and is there any different features of dividends between the markets.

From a different perspective from the above question, as an hypothesis, many excellent companies are located on megalopolises. On the other hand, there are several blue chip companies on provincial cities and these companies support local employments and citizens feel strong necessity for the companies. Therefore, its stockholders have been formed long-term position rather than speculators' behaviors and contributed to design and operate the corporate governance. Consequentially, the company's stock price has been stable and keeps resilient. These are hidden gems.

3. What local excellent companies were ranked in the index, where they are located on and how many employees contributed to generate turn-overs and profits for a company.

3 Dataset manipulation

There were two data sources; the one was yfinance and the other was yahooquery. The communication between the client server's requests and yfinance servers' responses, such as info.get, were executed one-by-one on each requests and needed to take several minutes to gather a chunk of companies' data (e.g. over 500 companies for S&P 500). In addition, yfinance was restricted to obtain market capitalization data because it

can provide only share price and number of issued stocks but it contains only addition-and-reduction of issued stocks. The analyst should calculate the market capitalization through complex data gathering processes by themselves. On the other hands, yahooquery stores market capitalization at a single point of time and equip download function, which execute a batch processing of summarized data. The function contributed to shorten the processing time for data gathering processes. In this project, yahooquery was mainly used for the analysis rather than yfinance.

In the view of eliminating servers load, downloaded the data gathered from yahooquery in JSON files and short-cutting data gathering time for executing processes.

Demonstrated whole data manipulation procedures in the attached notebook file `DSM050_MidtermCW.ipynb`.

4 Exploratory Data Analysis and Visualization

4.1 US market

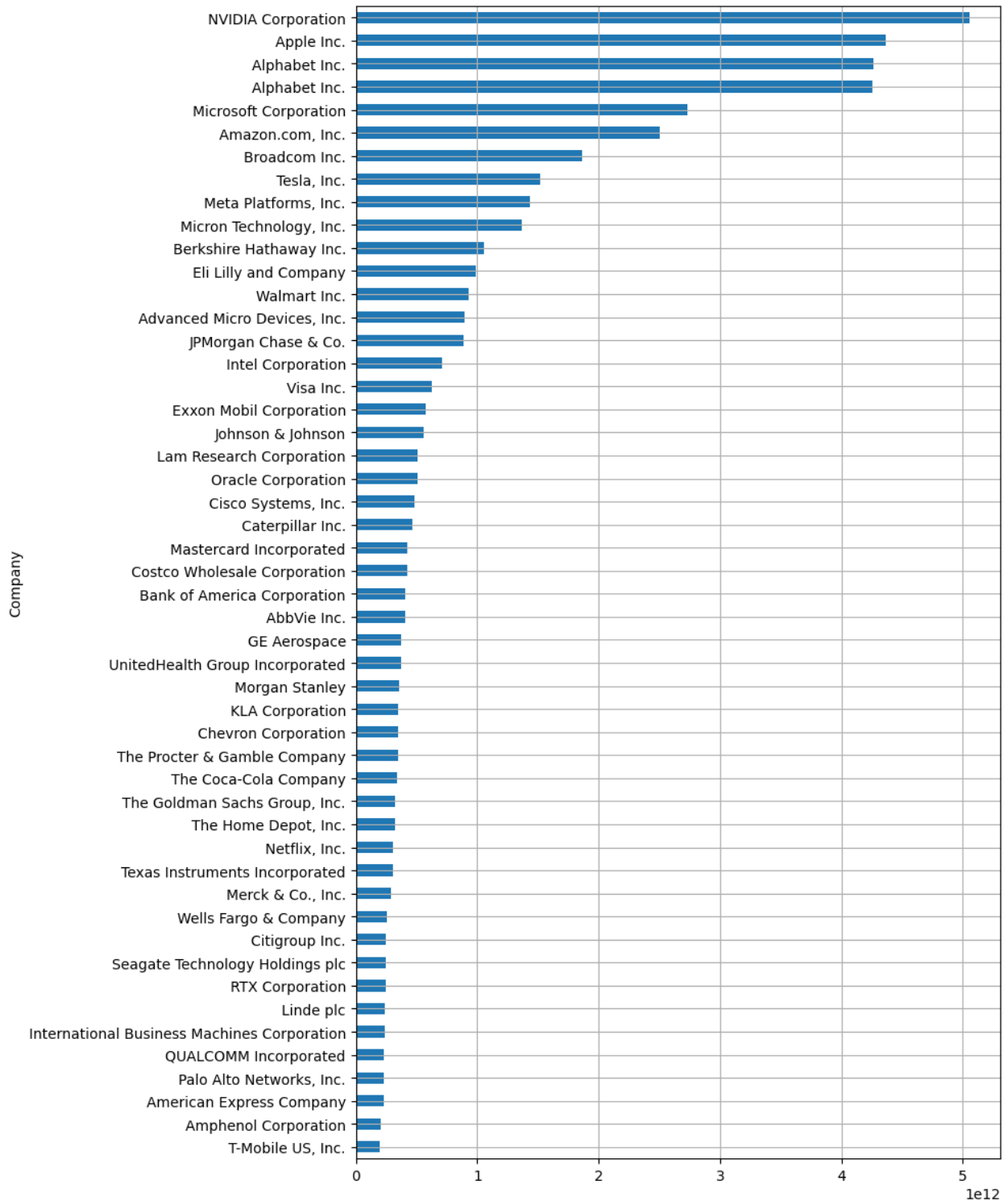


Figure 1: S&P 500 Market Capitalization as of June 23, 2026

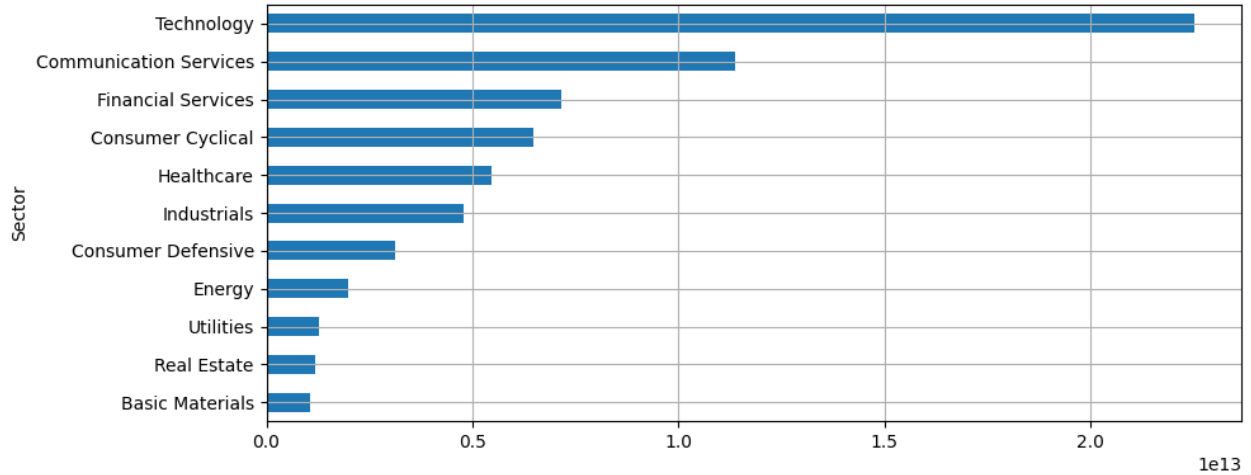


Figure 2: S&P 500 Market Capitalization by sectors as of June 23, 2026

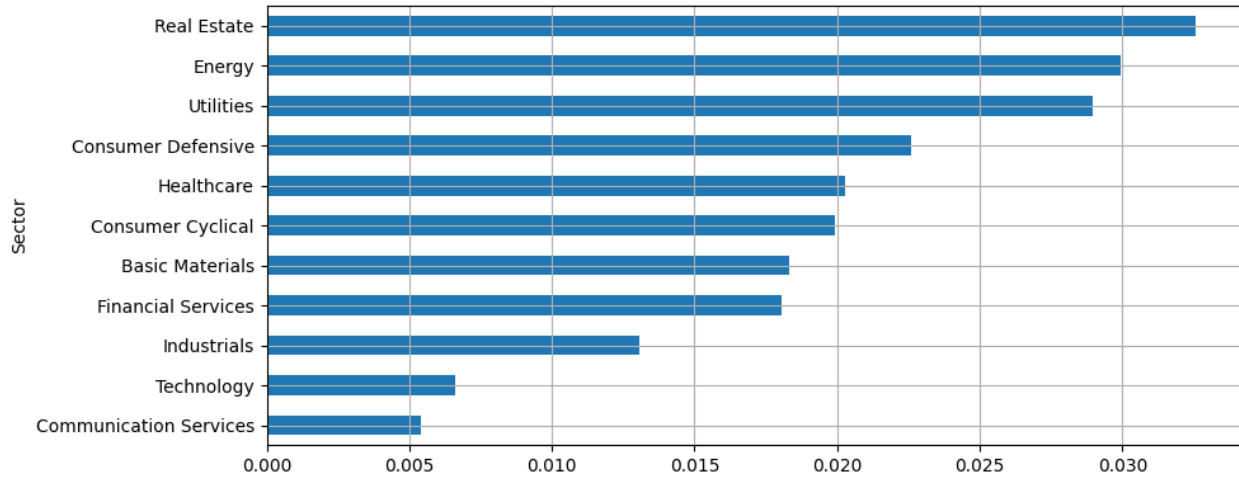


Figure 3: S&P 500 Dividend Yield by sectors as of June 23, 2026

In the US market, technology companies were quite huge worth and 7 companies, such as NVIDIA, Apple, Alphabet, Microsoft, Amazon, Broadcom and Micron, were over one trillion USD of market capitalization. Interestingly, three companies in the trillion-club were semiconductor companies. They have dominated inside chips in a computer and met demands of high-speed processing for consumers over the world. These roles were favored for traders. The technology sector was by far the best industry in the US market and the communication services sector, such as Meta and Netflix, was the second place and these sectors were close to each other because communication service have served on technology platforms, the platforms were depending on the technology. Even though two sectors' the market capitalization were high, its dividend yields were relatively low, approximately 0.5%. It implies traders in US market accepted low dividends but they need market capitalization and stock price growth.

4.2 UK market

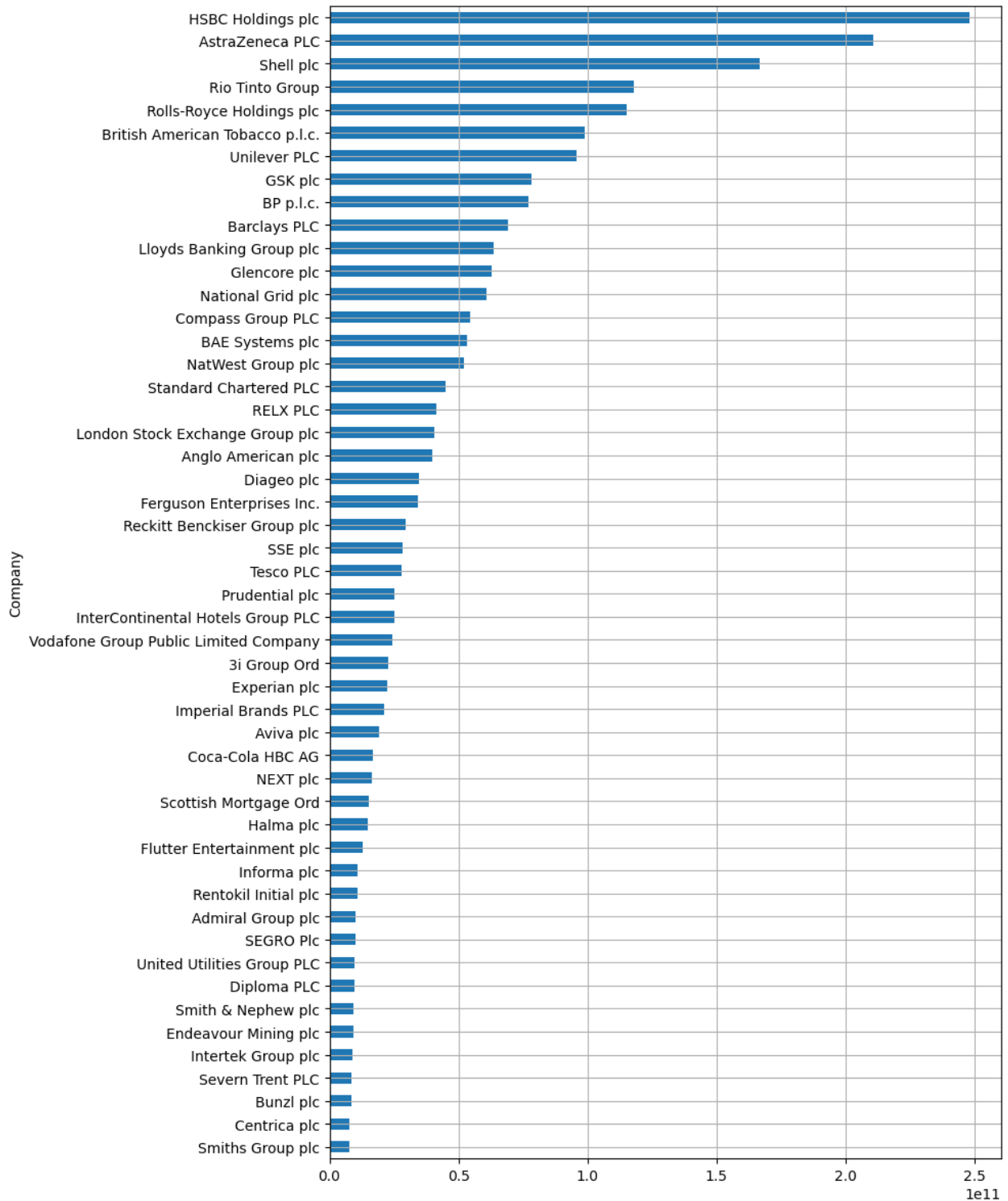


Figure 4: FTSE 100 Market Capitalization as of June 23, 2026

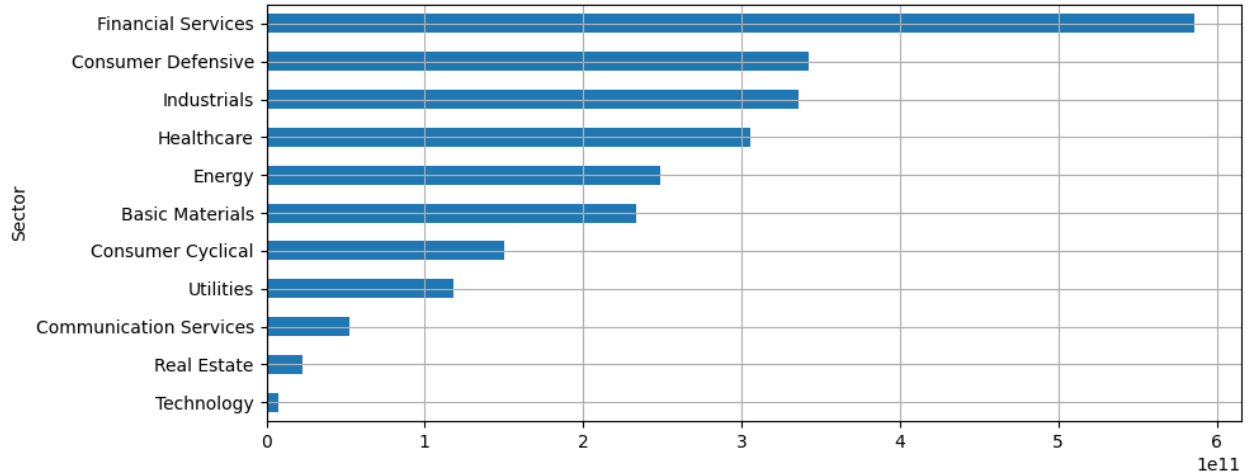


Figure 5: FTSE 100 Market Capitalization by sectors as of June 23, 2026

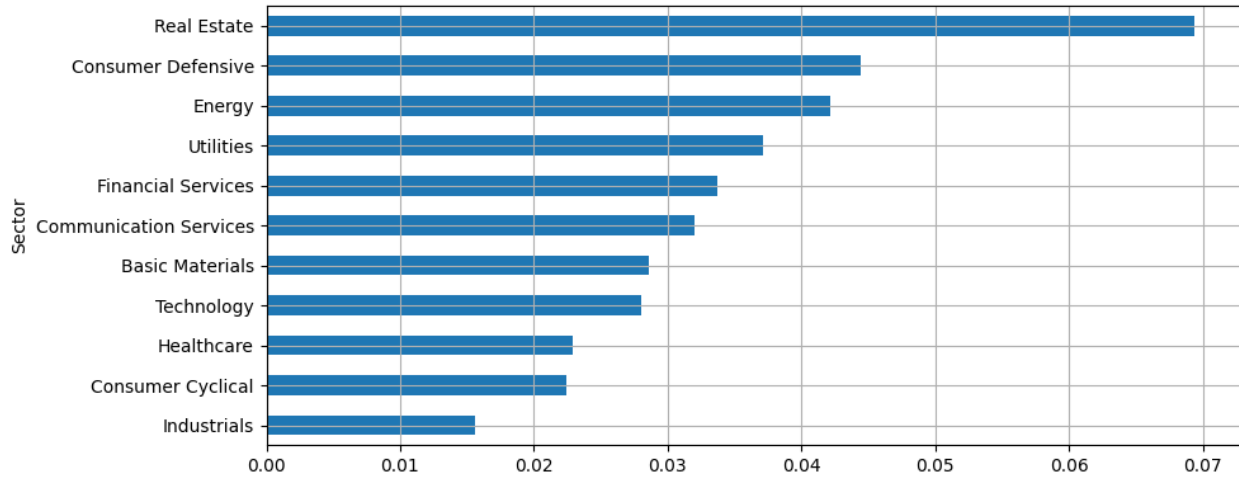


Figure 6: FTSE 100 Dividend Yield by sectors as of June 23, 2026

Top-ranked companies in FTSE 100 were relatively smaller than those of S&P 500. The market capitalization of HSBC, the highest worth company in FTSE 100, was around 250 billion GBP and the finance sector's value was around 600 billion GBP. On the other hand, there were 7 companies, whose values were over 1 trillion USD, in S&P 500. In addition, financial services, consumer products, industrials and healthcare, called as old-economy and defensive sectors by traders were relatively stronger than technology sector, which was the highest sector in S&P 500. And there were 6 sectors whose dividend yield were over 3%, surprisingly real estate sector was over 7%. The traders invested in FTSE 100 companies prefers a company, whose financial performance was relatively stable and pay high-yield dividends in annual.

4.3 Japan market

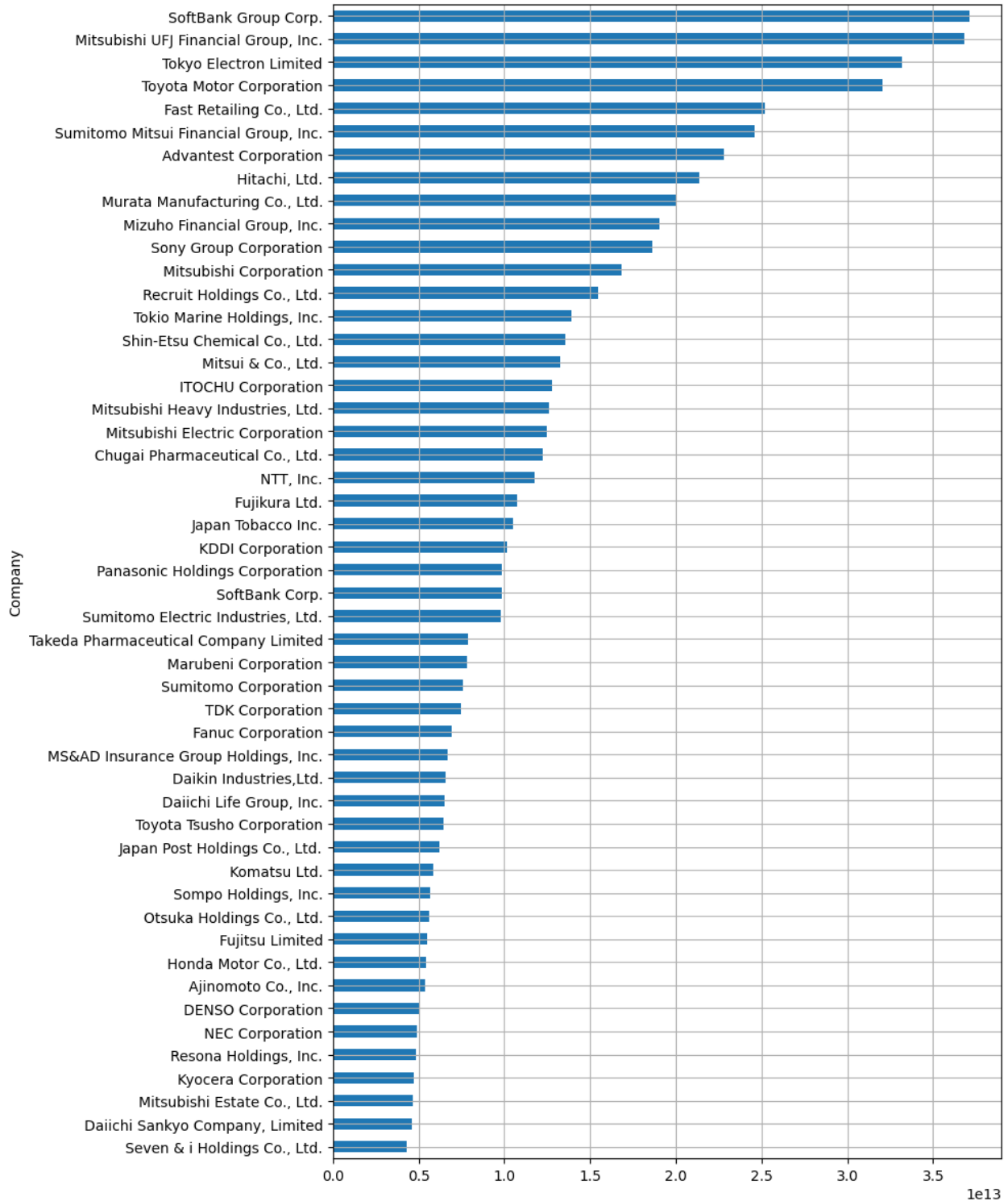


Figure 7: Nikkei 225 Market Capitalization as of June 23, 2026

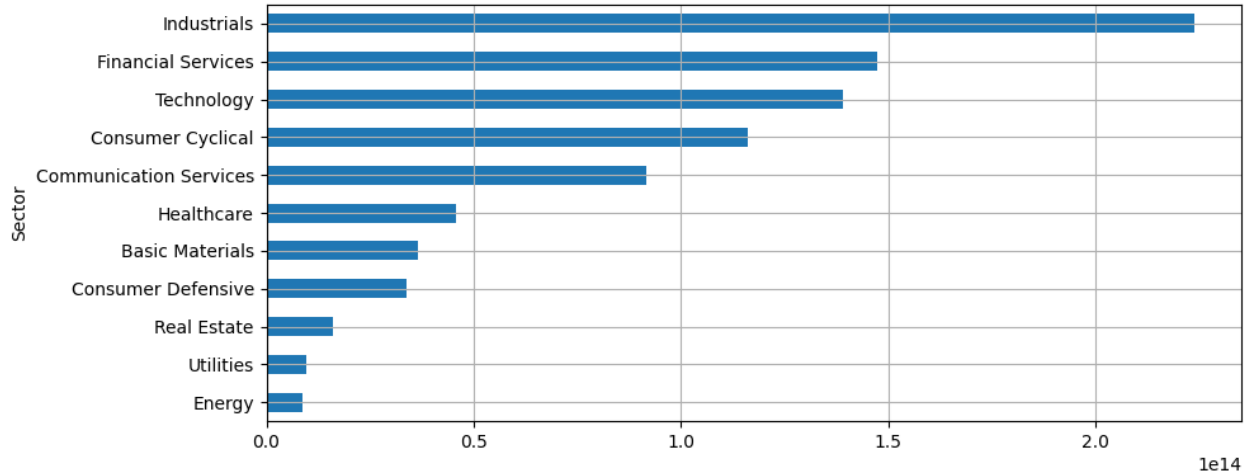


Figure 8: Nikkei 225 Market Capitalization by sectors as of June 23, 2026

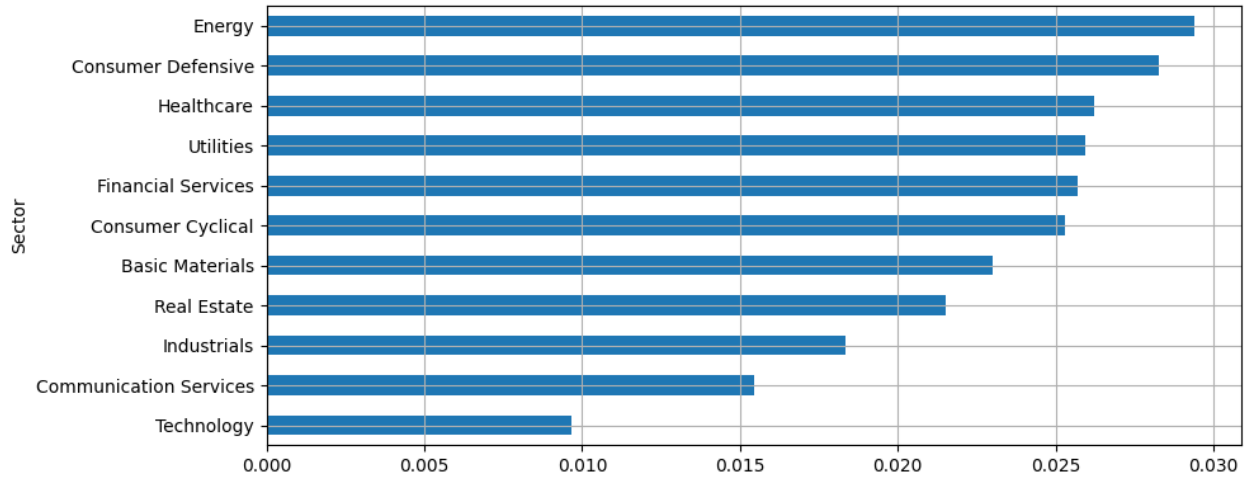


Figure 9: Nikkei 225 Dividend Yield by sectors as of June 23, 2026

SoftBank Group was the highest value in Nikkei 225. The group owns major telecommunication company, which acquired Vodafone Japan at 2006, and turned their strategy to more focusing on investment businesses. They invested technology companies, such as Arm and OpenAI, gains on valuation of these securities were contributed their worth. Even though technology sector was highlighted these days, industrial sector was the strongest sector in Japan. Toyota, Honda and Hitachi were national brands and semiconductor related products, such as materials, Shin-Etsu Chemical, and manufacturing machines, Tokyo Electron and Advantest, established strong presences. Industrial sector's market capitalization was over 200 trillion JPY. In dividend yields in Nikkei 225, all sectors except for technology sector were 1.5% to 3%, which was well-balanced level rather than S&P 500, low dividends, and FTSE 100, high dividends. The trend in Japanese stock market was favoring technology and chip companies such as Softbank group and semiconductors, but industrial sector was still strong and those of companies build its exclusive brands over a long period of time. If a trader seek short-term capital gain, technology companies are targets for trading. On the other hand, seeking long-term income gain, industrial companies are suitable.

5 Declaration of AI Tools

To use of AI tools properly, needed to understand the policy of the assignment and instruction of prohibited usage, which are generating full analytical reports, writing entire sections of the final report and fabricating data for analysis.

Within the allowable usage, AI Tools were used as below during preparation periods from 8 June, 2026 to 13 July, 2026:

- Gather relative academic reports from world-wide web with discussing the intention of the reports to Google Gemini.
- Generate stock indexes lists for S&P500, FTSE100 and Nikkei by using Google Gemini and ChatGPT.
- Asking to Google Gemini for preparing python code when obstacles and errors were appeared.

The differentiations between the author and AI tools; the author should design whole story of the report with definite intention occurred from thoughts and experiences. On the other hands, AI tools support the author's activity to gather information, simple and repetitive tasks and python coding. All outputs suggested by the AI tools were verified by the author checking its citation, do not simply paste it. Kept paying skepticism.

Even though keeping the above roles, blending suggestions from browser-embedded AI tools, such as Chrome AI mode, to the author's thought because searching the past research and cutting-edge ideas on the web was an essential action for designing the upstream processes of the report (E.g. aim, object and research questions) . AI tools could expand the author's ability but the core of the original intention and the strategy should be produced by myself.

6 Dashboard

To find local excellent companies from each index, a dashboard is an useful tool for viewing geographical location with plotting its scale indicator, such as number of employees. An user can zoom the area shown by the vertical bar graph and find company information by placing cursor over.

Walmart in Arkansas, Home Depot in Atlanta, FedEx in Memphis, Target in Minneapolis and Berkshire Hathaway in Omaha, these companies might be local excellent companies in US.

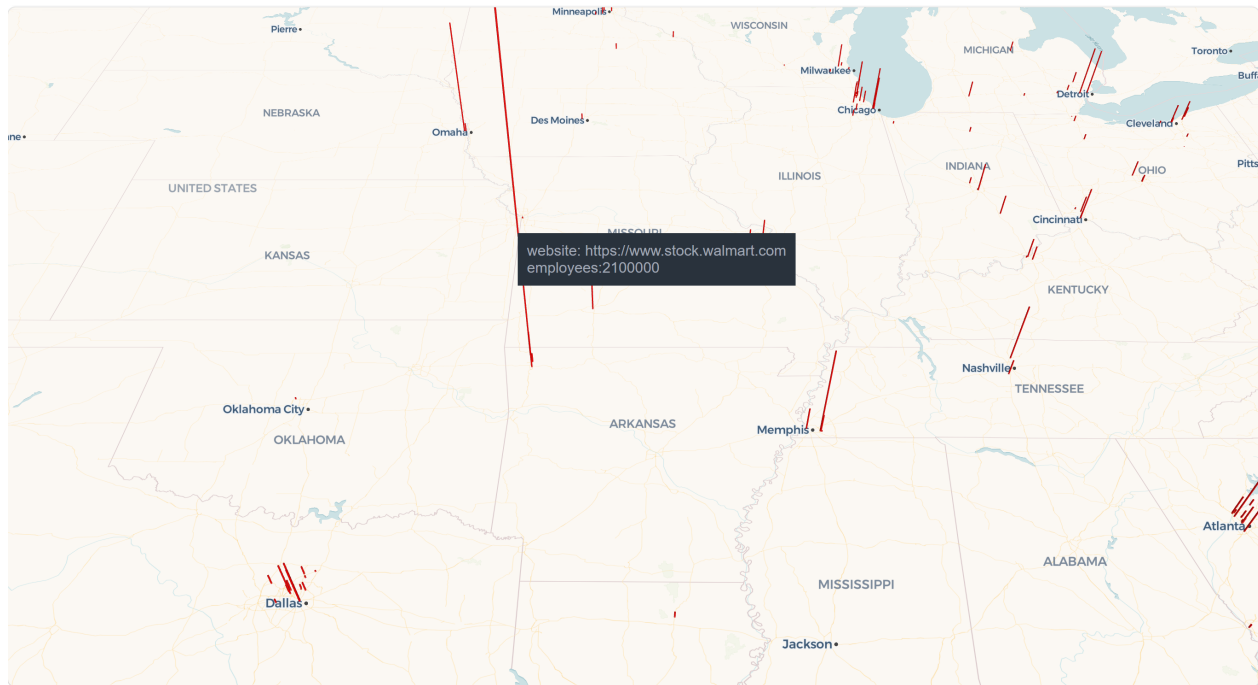


Figure 10: Number of employees of S&P 500 companies

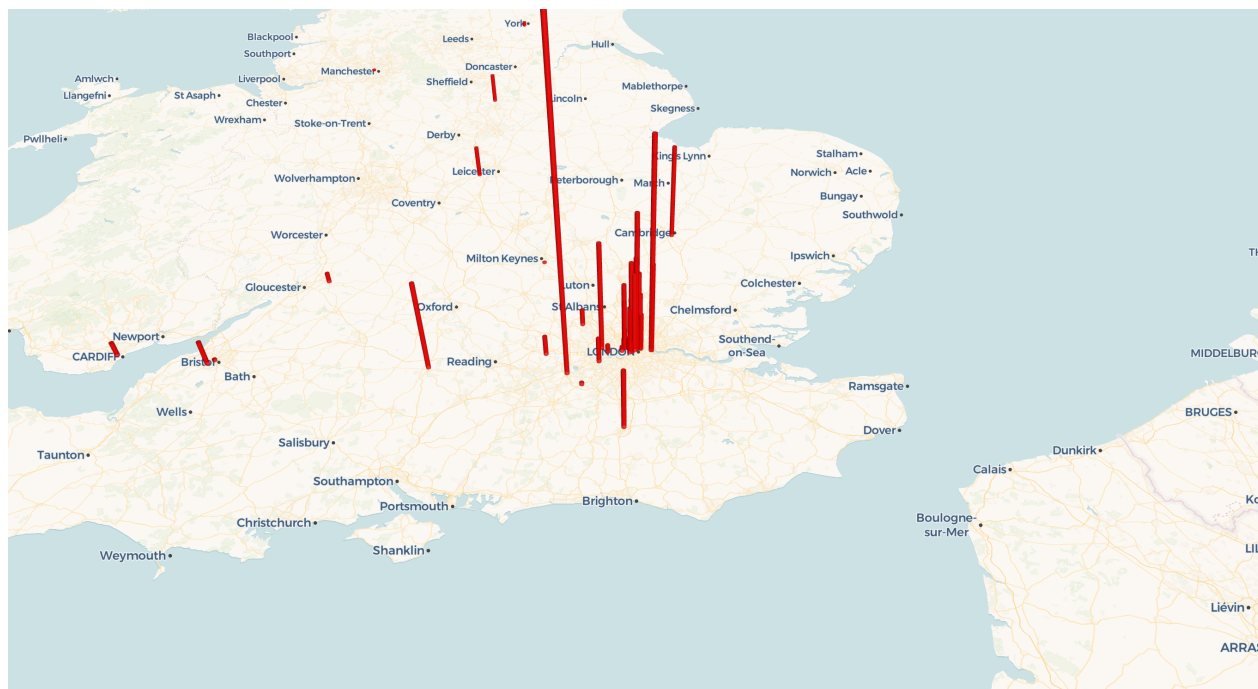


Figure 11: Number of employees of FTSE 100 companies

AstraZeneca in Cambridge, Admiral Group in Cardiff, Frasers in Derbyshire, Next in Leicester and Imperial Brands in Bristol might be local excellent companies in UK.

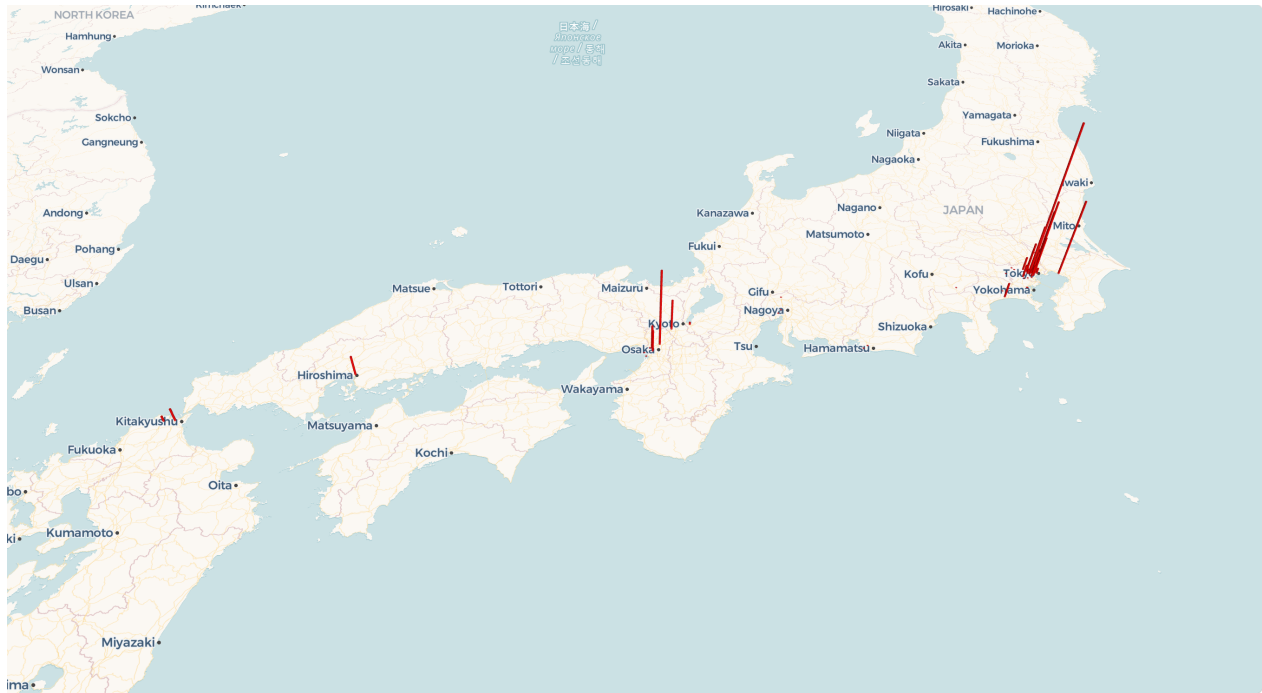


Figure 12: Number of employees of Nikkei 225 companies

Panasonic in Osaka, Murata in Kyoto, Mazda in Hiroshima, Yaskawa and ToTo in Fukuoka, these companies might be local excellent companies in Japan.

To verify whether these listed-companies are excellent or not, reading through their reading disclosure documents, such as 10-K and annual reports and evaluating financial performances.

7 Interpretation and Discussion

7.1 Insights from the research

According to the research results, the market capitalization of top S&P 500 companies were significant higher than the other companies in FTSE 100 and Nikkei 225. Technology companies, especially semiconductor and semiconductor related products companies were favored in S&P 500 and Nikkei 225. On the other hand, durable companies in defensive industries were still strong in FTSE 100. If an investor seek to earn capital gain, selecting S&P 500 companies. However, they are preferred to earn income gain, FTSE 100 companies were better choice. Nikkei 225 companies were a balanced option for both sides. By developing the dashboard, investors could identify the geographically distribution in the country and find local excellent companies. In fact, there are several companies, whose head-quarter is in a provincial city but their business covers around the world or keep dominant presence in the market.

7.2 Limitation

The stock market was like a vast ocean and it was difficult to find definitive patterns for supporting investor's decision. The research was mainly focusing the current status of investor's evaluation, which was represented in the stock price and market capitalization, with considering their dividends yield, and was not enough for verifying companies financial information, such as profit-and-loss statements, balance sheets and cash-flow statements. In addition, geographical investigation could find the simple distribution in the countries and successfully plotted in maps but the reason why the provincial city attracted companies, why companies are

still live there and are there any sector clusters, were not mentioned in the research. These topics are needed to study further more.

8 References

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